TITLE: Phan Tuấn Kiệt 2352654 Date: HW2

Activity 1:

Three node (12,0) ; (9,3) ; (6;6)

Sort these three node from lower to higher: (6;6) < (9;3) < (12;0)

First of all node (12;0) have the higher impurity because In this node, all values concentrated in a single class. While in node (6;6), the distribution of each class are equal which mean that we have 50% chance of predicting correctly. The node (9;3) if we take a value from this node and guess the value belong to class 1, we have 75% correctly so that it have more impurity than node (6;6).

node (12,0) $p_1 = \frac{12}{12} = 1$; $p_2 = \frac{0}{12} = 0$

$H((12,0)) = -(1.\log_2 1 + 0.\log_2 0) = 0$

=> no entropy in this node

node (9,3) $p_1 = \frac{9}{12} = 0,75$; $p_2 = \frac{3}{12} = 0,25$

$H((9,3)) = -\left(\frac{9}{12}.\log_2\frac{9}{12} + \frac{3}{12}.\log_2\frac{3}{12}\right) \approx 0,8113$

=> entropy of this nod quite high

node (6,6) $p_1 = \frac{6}{12} = 0,5$; $p_2 = \frac{6}{12} = 0,5$

$H((6,6)) = -(0,5.\log_2 0,5 + 0,5.\log_2 0,5) = 1$

=> entropy of this node reaching the higher value.

Activity 2:

S1 split into 2 children A(6P, 1N) ; B(5P, 6N)

Gini impurity (A) $= 1 - \left(\frac{6}{7}\right)^2 - \left(\frac{1}{7}\right)^2 \approx 0,2449$

Gini impurity (B) $= 1 - \left(\frac{5}{11}\right)^2 - \left(\frac{6}{11}\right)^2 \approx 0,4959$

Gini Index (S_1) $= \frac{7}{18}.0,2449 + \frac{11}{18}.0,4959 \approx 0,3983$

Just write it

Được quét bằng CamScanner

TITLE: Date:

S2 split into 2 children C(8P,4N) ; D(3P,3N)

Gini impurity (C) = $1-\left(\frac{8}{12}\right)^2-\left(\frac{4}{12}\right)^2 \approx 0,444$

Gini impurity (D) = $1-\left(\frac{3}{6}\right)^2-\left(\frac{3}{6}\right)^2 = 0,5$

Gini index (S_2) = $\frac{12}{18} \cdot 0,4444 + \frac{6}{18} \cdot 0,5 \approx 0,4629$

In CART it prefer to choose the split produces that have the smaller gini value so that it will choose S_1 for the split produces.

Choosing split produces which have smaller Gini value mean that split the data such that the class distribution within each child branch is skewed toward one side of the class range. As a result, the accuracy of the leaf will increase because there will be one class with a higher allocation percentage than the others

Activity 3:

values	4	7	9	13	18	21	29
label	Y	Y	Y	N	N	N	Y

three plausible cut points are : 8 ; 11 ; 19

8
- <8 [Y:2, N:0] ⇒ Gini impurity = $1-\frac{2^2}{2}-\frac{0^2}{2}=0$
- >=8 [Y:2, N:3] ⇒ Gini impurity = $1-\frac{3^2}{5}-\frac{2^2}{5}=0,48$

⇒ Gini index = $0,45 \cdot \frac{5}{7} \approx 0,3429$

11
- <11 [Y:3, N:0] ⇒ Gini impurity = $1-\frac{3^2}{3}-\frac{0^2}{3}=0$
- >=11 [Y:1, N:3] ⇒ Gini impurity = $1-\frac{1^2}{4}-\frac{3^2}{4}=0,375$

⇒ Gini index = $0,375 \cdot \frac{4}{7} \approx 0,2143$

Just write it

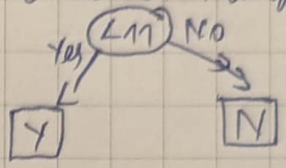
Được quét bằng CamScanner

TITLE: Date:

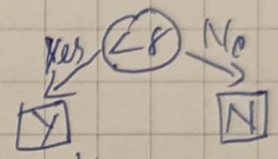
19
- <19 — Y: 3, N: 2 ⇒ Gini impurity = $1 - \frac{3}{5}^2 - \frac{2}{5}^2 = 0,48$
- >19 — Y: 1, N: 1 ⇒ Gini impurity = $1 - \frac{1}{2}^2 - \frac{1}{2}^2 = 0,5$

⇒ Gini index = $0,45 \cdot \frac{5}{7} + 0,5 \cdot \frac{2}{2} \approx 0,4857$.

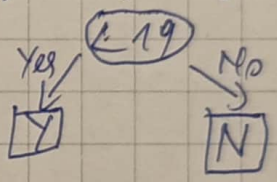
cut point 11 have the smaller gini value so I choose it for split produce.



Specifically, ~~at~~ using cut point 11, ~~if~~ ~~gini value~~ according the dataset, the left range of 11 will yield a result that is 100% classified as class Y will Over 11 range will give 75% accuracy for classify N class.



cut point 8; the left range also give 100% accuracy but the right range ~~gi~~ only give 60% ~~accuracy~~ correctly for classify class N ~~w~~which is lower than the previous cut points.



the accuracy of left ~~side~~ range and right range of cut point 19 are 60% for class Y on left range and 50% for class N on right side, respectively. It evidence that this cut-off point carries a higher risk of misclassification compared to the two cut-off points above.

Activity 4:

- follow reduced – error pruning help reduce so we choose prune. The purpose of split tree that increase the accuracy of decision, but in this case, split leading to the bad result than prune so choose prune.
- Some time, there are some sample lake of data. In this case, there are two way to deal this situation.
  + Choose the most common value.
    - for categorical feature, we ~~chose~~ select the most comon value to fill in the missing entries.
    - for numerical feature, we seler the mean or median value instead.

Pros: easy to build

Cons: This sample is affected bay other data samples.

Just write it

Được quét bằng CamScanner

TITLE: Date:

+ ~~Use~~ fill missing values based on data from another attribute.
  - Categorical feature will choose ~~the~~ another feature that have more ~~similar value~~ correlation with remaining feature and use it as a guide
  - Numerical feature: use choose ~~the~~ another feature that have ~~relation~~ relationship with to build a linear regression to predict the missing value

Pros: ensure that the missing samples still conform to the data distribution pattern.

Cons: hard to build and making the training process take longer.

Activity 5:

Use seed = 31 for split data
Use seed = 31 for DecisionTree classifier
+ Maximum depth: 20
+ Number of leaves: 139
+ Training score : 1.0
+ Validation score: 0,8107
+ Top 3 features: column 558; 86; 9

I will change Maximum depth into 15.
First of all, reduce the maximum dept tree leading to reduce the number of leaves. ~~Reduce height~~ The optimize depth of tree is 20, when reduce the maximum dept of tree into 15 means the accuracy of training also reduce to. The maximum depth be reduced so that the model become simpler means this model can get underfitting. The ~~value~~ validation score also reduced by reducing depth tree. The top 3 features will be unchanges because these features are ~~change~~ chosen for root and the child of root.

After change the value of max depth into 15 and run code:
+ Maximum depth: 15
+ Number of leaves: 129
+ training score: 0,9988
+ Validation score: 0,8235
+ Top 3 features: column 558; 86; 9

Just write it

Được quét bằng CamScanner

TITLE: Date:

number of leaves and training score reduce as predicted. However the value of validation score increase, maybe when reduce max depth make the model reduce overfitting phenomenon leading to increase validation score. Top 3 feature still remain stable as the previous run.

Just write it

Được quét bằng CamScanner